Question Number : 51 Question Id : 6952782186 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An oxide of iron contains 69.9% iron, its empirical formula, is:

(Given : Molar mass of Fe and O are 56 and 16 g mol^{-1} respectively.)

Options :

6952787666. FeO

6952787667. Fe_2O_3

6952787668. Fe_3O_4

6952787669. FeO_3

Question Number : 52 Question Id : 6952782187 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If shortest wavelength of hydrogen atom in Lyman series is x, then longest wavelength in Balmer series of He^+ is:

Options :

6952787670. $\frac{9x}{5}$

6952787671. $\frac{36x}{5}$

6952787672. $\frac{x}{4}$

6952787673. $\frac{5x}{9}$

Question Number : 53 Question Id : 6952782188 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

	List-I Orbital		**List-II Radial nodes and nodal plane**
A.	2s	I.	1 Radial node + two nodal planes
B.	3s	II.	1 Radial node + one nodal plane
C.	3p	III.	2 Radial nodes + No nodal plane
D.	4d	IV.	1 Radial node + No nodal plane

Choose the ***correct*** answer from the options given below:

Options :

6952787674. A-IV, B-I, C-III, D-II

6952787675. A-IV, B-II, C-III, D-I

6952787676. A-III, B-I, C-IV, D-II

6952787677. A-IV, B-III, C-II, D-I

Question Number : 54 Question Id : 6952782189 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The pairs among
$A = \left[SO_3^{2-}, CO_3^{2-} \right]$, $B = \left[O_2^{2-}, F_2 \right]$, $C = \left[CN^-, CO \right]$, $D = \left[NH_3, H_3O^+ \right]$ and
$E = \left[MnO_4^{2-}, CrO_4^{2-} \right]$ that do not have similar Lewis dot structure are

Options :

6952787678. A, B and E

6952787679. A and E

6952787680. B, C and D

6952787681. C and D

Question Number : 55 Question Id : 6952782190 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Arrange the following isothermal processes in order of the magnitude of the work (p − V) involved between states 1 and 2.

A. Expansion in single stage w_A
B. Expansion in multi stages w_B
C. Compression in single stage w_C
D. Compression in multi stages w_D

Choose the correct option.

Options :

6952787682. $|w_B| > |w_A| > |w_C| > |w_D|$

6952787683. $|w_C| > |w_D| > |w_A| > |w_B|$

6952787684. $|w_C| > |w_D| > |w_B| > |w_A|$

6952787685. $|w_B| > |w_A| > |w_D| > |w_C|$

Question Number : 56 Question Id : 6952782191 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

When 0.25 moles of a non-volatile, non-ionizable solute was dissolved in 1 mole of a solvent the vapor pressure of solution was *x* % of vapor pressure of pure solvent. What is *x* %?

Options :

6952787686. 50%

6952787687. 60%

6952787688. 70%

6952787689. 80%

Question Number : 57 Question Id : 6952782192 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

One mole each of He and A(g) are taken in a 10 L closed flask and heated to 400 K to establish the following equilibrium.

$A(g) \rightleftharpoons B(g)$

K_c for this reaction at 400 K is 4.0. The partial pressures (in atm) of He and B(g) are respectively (at equilibrium)

(Assume He, A(g) and B(g) behave as ideal gases)

(Given : R = 0.082 L atm K^{-1} mol^{-1})

Options :

6952787690. 3.28, 2.624

6952787691. 2.624, 3.28

6952787692. 3.28, 0.656

6952787693. 0.656, 6.56

Question Number : 58 Question Id : 6952782193 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following data.

Electrolyte	Λ°_m (S cm^2 mol^{-1})
$BaCl_2$	x_1
H_2SO_4	x_2
HCl	x_3

$BaSO_4$ is sparingly soluble in water. If the conductivity of the saturated $BaSO_4$ solution is x S cm^{-1} then the solubility product of $BaSO_4$ can be given as

(Here $\Lambda_m = \Lambda^\circ_m$)

Options :

6952787694. $\frac{10^6 x^2}{\alpha^2 \left(x_1 + x_2 - 2x_3\right)^2}$

6952787695. $\frac{x^2}{\left(x_1 + x_2 - 2x_3\right)^2}$

6952787696. $\frac{\alpha^2 \left(x_1 + x_2 - 2x_3\right)^2}{10^6 x^2}$

6952787697. $\frac{x^2}{\left(x_1 + x_2 + 2x_3\right)^2}$

Question Number : 59 Question Id : 6952782194 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: Aluminium is more electropositive than thallium as the standard electrode potential value of $E^\circ_{Al^{3+}/Al}$ is negative and $E^\circ_{Tl^{3+}/Tl}$ is positive.

Statement II: The sum of first three ionization enthalpies of boron is very high when compared to that of aluminium. Due to this reason boron forms covalent compounds only and aluminium forms Al^{3+} ion.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952787698. Both Statement I and Statement II are true

6952787699. Both Statement I and Statement II are false

6952787700. Statement I is true but Statement II is false

6952787701. Statement I is false but Statement II is true

Question Number : 60 Question Id : 6952782195 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The **correct** statements among the following are.

A. Basic vanadium oxide is used in the manufacture of H_2SO_4.
B. The spin-only magnetic moment value of the transition metal halide employed in Ziegler-Natta polymerization is 2.84 BM.
C. The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.
D. The number of electrons present in the outer most 'd' orbital of metal halide employed in Wacker process is 8.

Choose the correct answer from the options given below:

Options :

6952787702. A and B Only

6952787703. A, C and D Only

6952787704. C and D Only

6952787705. B, C and D Only

Question Number : 61 Question Id : 6952782196 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

List-I Electronic configuration of tetrahedral metal ion		List-II Crystal Field Stabilization Energy (Δ_t)	
A.	d^2	I.	− 0.6
B.	d^4	II.	− 0.8
C.	d^6	III.	− 1.2
D.	d^8	IV.	− 0.4

Choose the ***correct*** answer from the options given below:

Options :

6952787706. A-III, B-IV, C-II, D-I

6952787707. A-III, B-I, C-IV, D-II

6952787708. A-III, B-IV, C-I, D-II

6952787709. A-II, B-I, C-IV, D-III

Question Number : 62 Question Id : 6952782197 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following are true about the energy of the given d-orbitals of a tetrahedral complex?

A. $d_{xy} = d_{xz} > d_{x^2-y^2}$
B. $d_{xy} = d_{yz} > d_{z^2}$
C. $d_{x^2-y^2} > d_{z^2} > d_{xz}$
D. $d_{x^2-y^2} = d_{z^2} < d_{xz}$

Choose the correct answer from the given below:

Options :

6952787710. A, B and D only

6952787711. A and B only

6952787712. B and D only

6952787713. B, C and D only

Question Number : 63 Question Id : 6952782198 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

R_f value for 2-methylpropene in a solvent system (Ethyl acetate + ether) is 0.42.

2-methylpropene is treated with dilute H_2SO_4 to give major organic product (X).

R_f value for (X) in the same solvent system under identical condition will be:

Options :

6952787714. 0.42

6952787715. 0.82

6952787716. 0.62

6952787717. 0.12

Question Number : 64 Question Id : 6952782199 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: 2,6-diethylcyclohexanone and 6-methyl-2-n-propylcyclohexanone are metamers.

Statement II: 2,2,6,6 - tetramethylcyclohexanone exhibits keto-enol tautomerism.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952787718. Both Statement I and Statement II are true

6952787719. Both Statement I and Statement II are false

6952787720. Statement I is true but Statement II is false

6952787721. Statement I is false but Statement II is true

Question Number : 65 Question Id : 6952782200 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of CH_3MgBr with water.

Statement II: Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952787722. Both Statement I and Statement II are true

6952787723. Both Statement I and Statement II are false

6952787724. Statement I is true but Statement II is false

6952787725. Statement I is false but Statement II is true

Question Number : 66 Question Id : 6952782201 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: 3-phenylpropene reacts with HBr and gives secondary alkyl bromide having a chiral carbon atom as the major product.

Statement II: Aryl chlorides and aryl cyanides can be prepared by Sandmeyer reaction as well as Gattermann reaction.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952787726. Both Statement I and Statement II are true

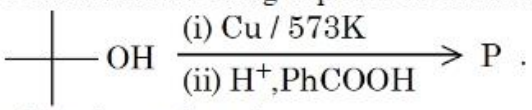
6952787727. Both Statement I and Statement II are false

6952787728. Statement I is true but Statement II is false

6952787729. Statement I is false but Statement II is true

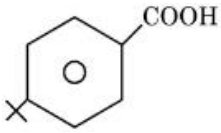
Question Number : 67 Question Id : 6952782202 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

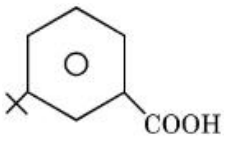
Consider the following sequence of reactions

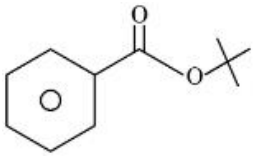


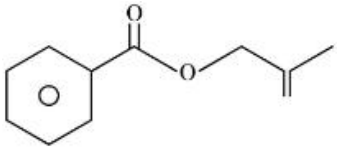
The major product P is:

Options :

6952787730. 

6952787731. 

6952787732. 

6952787733. 

Question Number : 68 Question Id : 6952782203 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Arrange the following compounds according to increasing order of boiling points.

n - C_4H_9OH (A), n - $C_4H_9NH_2$ (B), n - C_4H_{10} (C) and $C_2H_5NHC_2H_5$(D).

Options :

6952787734. C < B < A < D

6952787735. D < C < B < A

6952787736. C < D < B < A

6952787737. D < B < A < C

Question Number : 69 Question Id : 6952782204 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

	List-I Deficiency Disease		**List-II Vitamin**
A.	Scurvy	I.	Pyridoxine
B.	Convulsions	II.	Vitamin A
C.	Cheilosis	III.	Ascorbic Acid
D.	Xerophthalmia	IV.	Riboflavin

Choose the ***correct*** answer from the options given below:

Options :

6952787738. A-I, B-III, C-II, D-IV

6952787739. A-I, B-III, C-IV, D-II

6952787740. A-III, B-I, C-IV, D-II

6952787741. A-III, B-I, C-II, D-IV

Question Number : 70 Question Id : 6952782205 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match the **LIST-I** with **LIST-II**

	List-I Amino acid		**List-II Positive reaction/Test for functional group present in side chain of amino acid**
A.	Glutamine	I.	Hinsberg's test
B.	Lysine	II.	Neutral $FeCl_3$ test
C.	Tyrosine	III.	Ceric ammonium nitrate test
D.	Serine	IV.	Hoffman bromamide degradation

Choose the ***correct*** answer from the options given below:

Options :

6952787742. A-IV, B-II, C-I, D-III

6952787743. A-IV, B-I, C-II, D-III

6952787744. A-III, B-II, C-I, D-IV

6952787745. A-IV, B-I, C-III, D-II

Chemistry Section B

Section Id :	695278156
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	695278156
Question Shuffling Allowed :	Yes
Is Section Default? :	No

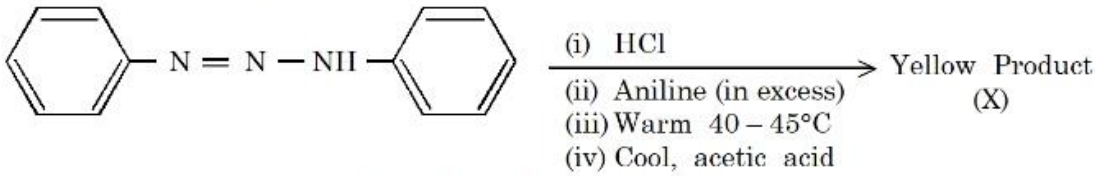
Question Number : 71 Question Id : 6952782206 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

First and second ionization enthalpies of lithium are 520 kJ mol^{-1} and 7297 kJ mol^{-1} respectively. Energy required to convert 3.5 mg lithium (g) into $Li^{2+}(g)$ [$Li(g) \rightarrow Li^{2+}(g)$] is _____ kJ mol^{-1} . (nearest integer)
[Molar mass of Li = 7 g mol^{-1}]

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 6952782207 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following sequence of reactions.



The percentage of nitrogen in the yellow product (X) formed is _____%.
(Nearest Integer)
(Given Molar mass in g mol^{-1} H:1, C:12, N:14)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 73 Question Id : 6952782208 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

4.7 g of phenol is heated with Zn to give product X. If this reaction goes to 60% completion then the number of moles of compound X formed will be
____ $\times 10^{-2}$.(Nearest Integer)
(Given molar mass in g mol^{-1} : H:1, C:12, O:16)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 6952782209 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Sucrose hydrolyses in acidic medium into glucose and fructose by first order rate law with $t_{1/2} = 3$ hour. The percentage of sucrose remaining after 6 hours is _____. (Nearest integer)
(Given : log 2 = 0.3010 and log 3 = 0.4771)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 75 Question Id : 6952782210 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the reaction $X \rightleftharpoons Y$ at 300 K. If ΔH^{θ} and K are 28.40 kJ mol^{-1} and 1.8×10^{-7} at the same temperature, then the magnitude of ΔS^{θ} for the reaction in J K^{-1} mol^{-1} is_____. (Nearest integer)
(Given : R = 8.3 J K^{-1} mol^{-1}, ln 10 = 2.3, log 3 = 0.48, log 2 = 0.30)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1